MATH 1271 **Name:** _____
Fall 2026
Quiz 10 - 10/15 Content
10/20/2026
Time Limit: 5 minutes **Student ID:** _____

This quiz consists of 1 questions across 2 pages worth 5 points.
Rules:

- You have the indicated amount of time above to complete the quiz and you must not begin working on the quiz until instructed to do so.
- You must show all of your work to receive full credit, unless stated otherwise. **On problems that require work to be shown, answers that are mysterious or unsupported will not receive full credit.**
- You may not use a calculator, textbook, notes or any other external resources.
- Please write your initials at the top of every page in the event the pages become separated.
- If you run out of space on a page, feel free to use the back of the page to continue your work.
- You understand that if you violate any of these or the University's rules, you will receive a score of zero on this quiz and you may be subject to further disciplinary action.

Following the University of Minnesota's policy on academic honesty, I pledge that the work shown on this quiz is entirely my own and that I have followed the rules stated above.

Signature: _____

Question	Points
1	5
Total:	5

1. (5 points) Differentiate the function:

$$h(t) = \sin(t^3 e^{-6t}). \tag{1}$$

You need not distribute any products across sums, or otherwise simplify any quantities computed after resolving all differentiation.

Solution:

$$\begin{aligned}
h'(t) &= \frac{d}{dt}(\sin(t^3 e^{-6t})) && (2)\\
&= (\frac{d}{dt}(t^3 e^{-6t})) \cdot \cos(t^3 e^{-6t}) && (3)\\
&= (3t^2 e^{-6t} + t^3 \cdot \frac{d}{dt}(e^{-6t})) \cdot \cos(t^3 e^{-6t}) && (4)\\
&= (3t^2 e^{-6t} + t^3 \cdot (\frac{d}{dt} - 6t) \cdot e^{-6t}) \cdot \cos(t^3 e^{-6t}) && (5)\\
&= (3t^2 e^{-6t} - 6t^3 e^{-6t}) \cos(t^3 e^{-6t}) && (6)
\end{aligned}$$